

DSBDA Lab Manual

310256 : Data Science and Big Data Analytics

Comprehensive Viva Questions & Answers

TE COMP (2019 Pattern) | Academic Year 2025-26 Sem-II

This document covers all possible viva questions from both Group A (Python/ML practicals) and Group B (Hadoop/MapReduce practicals) based on the lab manual theory.

GROUP A — Python & Machine Learning Practicals

Practical 1 & 2 — Data Wrangling

Q. What is Data Wrangling?

Ans: Data wrangling is the process of cleaning, organizing, and transforming raw data into a usable format for analysis. It involves mapping data from one form to another to handle large and complex datasets effectively.

Q. What is another name for Data Wrangling?

Ans: Data wrangling is also known as data cleaning or data munging.

Q. What are the goals of Data Wrangling?

Ans: Goals include: collecting data from multiple sources, providing accurate data for analysis, reducing time on preparation, allowing analysts to focus on analysis, and supporting better decision-making.

Q. What are the key steps in Data Wrangling?

Ans: Key steps are: (1) Discovery — understanding the data, (2) Structuring — organizing raw data, (3) Cleaning — fixing missing/inconsistent values, (4) Enriching — adding useful data, (5) Validating — verifying data quality, (6) Publishing — delivering the clean data.

Q. Which Python libraries are used for Data Wrangling?

Ans: Pandas and NumPy are the primary libraries used for data wrangling in Python.

Q. What is the purpose of pandas isnull() function?

Ans: isnull() is used to detect missing values in a DataFrame. It returns a Boolean DataFrame of the same shape, with True where values are missing (NaN).

Q. What does the describe() function do in pandas?

Ans: describe() generates summary statistics of numeric columns including count, mean, standard deviation, min, 25th/50th/75th percentiles, and max.

Q. What are outliers and how do you handle them?

Ans: Outliers are data points that deviate significantly from other observations. They can be handled by removing them, replacing with mean/median, using IQR-based filtering, or applying transformations like log scaling.

Q. What is data normalization?

Ans: Data normalization is the process of scaling numeric variables to a standard range (e.g., 0 to 1) so that no single variable dominates due to its magnitude.

Q. Why do we convert categorical variables to numeric?

Ans: Machine learning algorithms require numeric input. Categorical variables must be encoded (e.g., via Label Encoding or One-Hot Encoding) before passing them to ML models.

Q. What data transformation techniques are commonly applied?

Ans: Common techniques include log transformation (to reduce skewness), min-max normalization, z-score standardization, and square-root transformation.

Q. What tools are used for Data Wrangling apart from Python?

Ans: Excel / Power Query, Tabula, and Google DataPrep are commonly used tools.

Practical 3 — Summary Statistics

Q. What is Mean and how is it calculated?

Ans: Mean is the average value of a dataset. Formula: $\text{Mean} = (\text{Sum of all values}) / (\text{Total number of values})$. Example: For data [36, 40, 32, 42, 30], $\text{Mean} = 180/5 = 36$.

Q. What is Median?

Ans: Median is the middle value of a sorted dataset. If there is an even number of values, it is the average of the two middle values. It is robust to outliers.

Q. What is Mode?

Ans: Mode is the most frequently occurring value in a dataset. A dataset can have no mode, one mode (unimodal), or multiple modes (multimodal).

Q. What is Standard Deviation?

Ans: Standard deviation measures how spread out data values are around the mean. It is the square root of variance. A low SD means data is clustered near the mean; a high SD means it is spread out.

Q. What is Variance?

Ans: Variance is the average of squared differences from the mean. $\text{Standard Deviation} = \sqrt{\text{Variance}}$.

Q. What is a Percentile?

Ans: A percentile indicates the value below which a given percentage of observations fall. E.g., the 75th percentile means 75% of data points are below that value.

Q. Which libraries are used for statistical analysis in Python?

Ans: Pandas, NumPy, and Seaborn are commonly used for statistical analysis and visualization.

Q. What is the difference between descriptive and inferential statistics?

Ans: Descriptive statistics summarize and describe data (mean, median, SD). Inferential statistics use sample data to make inferences or predictions about a population.

Q. What is IQR (Interquartile Range)?

Ans: $\text{IQR} = Q3 - Q1$. It represents the range of the middle 50% of data and is used to detect outliers: values below $Q1 - 1.5 \times \text{IQR}$ or above $Q3 + 1.5 \times \text{IQR}$ are considered outliers.

Practical 4 — Linear Regression

Q. What is Linear Regression?

Ans: Linear regression is a supervised ML algorithm that models the relationship between a dependent variable and one or more independent variables by fitting a straight line ($y = mx + c$) to the data.

Q. What is Simple Linear Regression?

Ans: Simple Linear Regression predicts a response using a single feature. It assumes a linear relationship between one independent variable (x) and one dependent variable (y).

Q. What is Multiple Linear Regression?

Ans: Multiple Linear Regression models the relationship between two or more independent features and a response variable. It is an extension of simple linear regression.

Q. What is the equation of a linear regression line?

Ans: $y = b_0 + b_1x_1 + b_2x_2 + \dots + b_nx_n$, where b_0 is the intercept and $b_1 \dots b_n$ are coefficients of the features.

Q. What are applications of Linear Regression?

Ans: Applications include: trend line prediction, financial risk analysis, biology (studying variable relationships), predicting housing prices, and sales forecasting.

Q. Which Python libraries are used for Linear Regression?

Ans: Pandas (data handling) and Scikit-learn (sklearn) for building and evaluating the regression model.

Q. What is the Boston Housing Dataset?

Ans: It is a classic dataset with 506 samples and 13 features (like crime rate, number of rooms) used to predict median house prices (MEDV). It is often used to demonstrate regression.

Q. What metrics evaluate a regression model?

Ans: Common metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R^2) score.

Q. What does R-squared mean?

Ans: R^2 (coefficient of determination) indicates how well the model fits the data. $R^2 = 1$ means perfect prediction; $R^2 = 0$ means the model explains none of the variance.

Q. What is the difference between regression and classification?

Ans: Regression predicts a continuous output (e.g., house price), while classification predicts a categorical/discrete output (e.g., spam or not spam).

Practical 5 — Logistic Regression

Q. What is Logistic Regression?

Ans: Logistic Regression is a supervised learning algorithm used for classification problems where the output is categorical (e.g., yes/no, 0/1). It predicts the probability of belonging to a class using the sigmoid function.

Q. How is Logistic Regression different from Linear Regression?

Ans: Linear Regression predicts continuous values, while Logistic Regression predicts probabilities for categorical outcomes. Logistic Regression uses the sigmoid function to map outputs between 0 and 1.

Q. What are the types of Logistic Regression?

Ans: Three types: (1) Binary — two outcomes (yes/no), (2) Multinomial — three or more categories, (3) Ordinal — ordered categories like ratings.

Q. What is the Sigmoid Function?

Ans: The sigmoid function maps any real-valued number to a value between 0 and 1. Formula: $\sigma(z) = 1 / (1 + e^{-z})$. It is used to convert the linear output to a probability.

Q. When should you use Logistic Regression?

Ans: Use Logistic Regression when the dependent variable is categorical, when there is low correlation between independent variables, fewer outliers, and data is not too sparse.

Q. What is the Social Network Ads dataset?

Ans: It contains information about users (age, salary) and whether they purchased a product based on a social network ad. It is used for binary classification.

Q. Which Python libraries are used for Logistic Regression?

Ans: Pandas (data handling), Scikit-learn (model), Seaborn and Matplotlib (visualization).

Q. What is a Confusion Matrix?

Ans: A confusion matrix summarizes the performance of a classification model showing True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).

Q. What is Accuracy in classification?

Ans: Accuracy = $(TP + TN) / (TP + TN + FP + FN)$. It measures the percentage of correctly classified instances.

Q. What is overfitting?

Ans: Overfitting occurs when a model learns the training data too well (including noise), resulting in poor performance on new/unseen data.

Practical 6 — Naïve Bayes Classifier**Q. What is the Naïve Bayes Classifier?**

Ans: Naïve Bayes is a supervised learning algorithm for classification based on Bayes' Theorem. It assumes all features are independent of each other (the 'naïve' assumption).

Q. What is Bayes' Theorem?

Ans: Bayes' Theorem: $P(A|B) = [P(B|A) * P(A)] / P(B)$. It describes the probability of an event based on prior knowledge of related conditions.

Q. Why is it called 'Naïve' Bayes?

Ans: It is called 'Naïve' because it naively assumes that all features/predictors are mutually independent, which is rarely true in real-world data.

Q. What are the types of Naïve Bayes classifiers?

Ans: Three types: (1) Bernoulli — binary data (True/False), (2) Multinomial — word frequency in text classification, (3) Gaussian — continuous data assuming normal distribution.

Q. What are the advantages of Naïve Bayes?

Ans: Fast and easy to implement, works well with small datasets, suitable for multi-class problems, and performs well with categorical data.

Q. What are the disadvantages of Naïve Bayes?

Ans: Assumes feature independence (not always true), cannot handle unseen categories well (zero frequency problem), and probability predictions may not always be accurate.

Q. What is the Zero Frequency Problem in Naïve Bayes?

Ans: If a category in the test set was not present in training, the classifier assigns zero probability, making prediction fail. This is solved using Laplace Smoothing (adding 1 to all counts).

Q. What are the applications of Naïve Bayes?

Ans: Spam email filtering (Gmail), sentiment analysis, real-time prediction, multi-class classification, and recommendation systems.

Q. What is the Iris dataset?

Ans: The Iris dataset has 150 samples with 4 features (sepal length, sepal width, petal length, petal width) and 3 classes (Iris setosa, versicolor, virginica). It is a benchmark classification dataset.

Q. What is a supervised learning algorithm?

Ans: A supervised learning algorithm learns from labelled training data where the correct output is provided. It learns a mapping from input to output and applies it to new data.

Practical 7 — NLP Text Preprocessing**Q. What is Data Preprocessing in NLP?**

Ans: Data preprocessing in NLP is cleaning and transforming raw text data into a structured format suitable for ML algorithms. It includes tokenization, stop word removal, stemming, and lemmatization.

Q. What is Tokenization?

Ans: Tokenization is the process of splitting text into smaller units called tokens — either words (word tokenization) or sentences (sentence tokenization).

Q. What are Stop Words?

Ans: Stop words are common words (like 'the', 'is', 'in', 'a') that carry little meaning and are removed during text preprocessing to reduce noise and processing time.

Q. What is Stemming?

Ans: Stemming reduces words to their root form by removing suffixes, even if the root is not a proper word. Example: 'liked', 'likes' → 'like'. It is fast but may produce non-meaningful roots.

Q. What is Lemmatization?

Ans: Lemmatization converts words to their base (dictionary) form while preserving meaning. Example: 'running' → 'run'. It is more accurate than stemming but slower.

Q. What is the difference between Stemming and Lemmatization?

Ans: Stemming is faster but may produce non-dictionary words (e.g., 'studies' → 'studi'). Lemmatization is slower but always produces valid words by considering context and grammar.

Q. What is Overstemming and Understemming?

Ans: Overstemming: different words are reduced to the same root (false grouping). Understemming: similar words remain different after stemming (missed grouping).

Q. What is POS Tagging?

Ans: Part-of-Speech (POS) tagging assigns grammatical labels (noun, verb, adjective, etc.) to each word in a sentence. It helps machines understand the role of each word.

Q. Which library is used for NLP in Python?

Ans: NLTK (Natural Language Toolkit) is the primary library. Sklearn and Pandas are also used for additional processing.

Q. What are applications of NLP preprocessing?

Ans: Applications include search engines, spam detection, sentiment analysis, chatbots, machine translation, and text summarization.

Practical 8 & 9 — Data Visualization (Titanic Dataset)

Q. What is Data Visualization?

Ans: Data visualization is the graphical representation of data using charts, graphs, and maps. It helps understand trends, patterns, and outliers easily, especially in Big Data analysis.

Q. What are the benefits of Data Visualization?

Ans: Benefits include: better analysis of performance data, faster decision-making, identifying patterns and trends, finding data errors, understanding stories in data, and gaining business insights.

Q. What is a Box Plot (Box and Whisker Plot)?

Ans: A box plot shows the statistical summary of data: minimum, Q1 (25%), median (50%), Q3 (75%), and maximum. The whiskers represent the range, and points beyond them are outliers.

Q. What does IQR represent in a box plot?

Ans: IQR (Interquartile Range) = $Q3 - Q1$. It represents the spread of the middle 50% of data and is shown as the box in the plot.

Q. What is the Titanic dataset?

Ans: The Titanic dataset has 891 rows with information about passengers (age, sex, class, fare, survival) who were aboard the Titanic. It is widely used for data analysis and classification.

Q. Which libraries are used for visualization in Python?

Ans: Seaborn and Matplotlib are the primary visualization libraries. Seaborn provides high-level statistical plots built on top of Matplotlib.

Q. What is Seaborn and how does it differ from Matplotlib?

Ans: Seaborn is a statistical visualization library that provides attractive default styles and built-in datasets. Matplotlib is lower-level and more customizable. Seaborn makes complex plots easier to create.

Q. What types of plots can be created with Seaborn?

Ans: Seaborn supports: box plots, violin plots, heatmaps, pair plots, bar plots, scatter plots, histogram (distplot), and count plots.

Q. What is the purpose of data visualization in Big Data?

Ans: Big Data is too large to interpret numerically. Visualization helps quickly identify patterns, anomalies, and trends in massive datasets, enabling data-driven decisions.

Practical 10 — Iris Dataset Visualization & Categorical Data

Q. What is Categorical Data?

Ans: Categorical data represents categories or labels. Types: Nominal (no order, e.g., colors, names) and Ordinal (ordered, e.g., ranks, grades).

Q. What are the formats of categorical data?

Ans: Raw Data (individual observations), Aggregated Data (counts per category), and Cross-tabulated Data (data in table form with rows and columns).

Q. What is a Histogram?

Ans: A histogram shows the frequency distribution of a continuous variable. The x-axis represents value ranges (bins) and the y-axis represents frequency/count.

Q. How do you identify outliers using a box plot?

Ans: Points that fall below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ are plotted as individual dots outside the whiskers of the box plot and are considered outliers.

Q. What is the difference between nominal and ordinal data?

Ans: Nominal data has no specific order (e.g., blood group A, B, O). Ordinal data has a defined order (e.g., satisfaction rating: low < medium < high).

GROUP B — Hadoop & MapReduce Practicals

Assignment 1 — Hadoop Word Count using MapReduce

Q. What is Hadoop?

Ans: Hadoop is an open-source framework for storing and processing large datasets in a distributed manner across clusters of commodity hardware. It uses HDFS for storage and MapReduce for processing.

Q. What is MapReduce?

Ans: MapReduce is a programming model and processing technique for distributed computing in Hadoop. It decomposes large jobs into parallel tasks. It consists of two steps: Map (converts data into key-value pairs) and Reduce (aggregates/combines the mapped data).

Q. What does the Map function do in MapReduce?

Ans: The Map function takes a set of data and converts it into another set of data, breaking individual elements into key-value tuples (Key, Value) pairs. Example in Word Count: each word is emitted as (word, 1).

Q. What does the Reduce function do in MapReduce?

Ans: The Reduce function takes the output from the Map step as input and combines those data tuples into a smaller set. Example: it sums all the 1s for each word key to get the total word count.

Q. What are the 5 steps in the MapReduce workflow?

Ans: The 5 steps are: (1) Splitting — input is split into chunks, (2) Mapping — each chunk is processed by a Mapper, (3) Intermediate Splitting (Shuffling) — key-value pairs are sorted and grouped, (4) Reducing — Reducer combines values for each key, (5) Combining — final output is produced.

Q. What is HDFS?

Ans: HDFS (Hadoop Distributed File System) is the storage component of Hadoop. It stores data across multiple nodes with replication (default 3 copies) for fault tolerance.

Q. What is a standalone setup in Hadoop?

Ans: In standalone (local) mode, Hadoop runs as a single Java process on one machine. It is used for debugging and testing MapReduce programs without a full cluster.

Q. What language is Hadoop's MapReduce written in?

Ans: Hadoop MapReduce programs are written in Java, though other languages can be used via Hadoop Streaming.

Q. What is the input and output of a Word Count MapReduce program?

Ans: Input: a text file with words. Map Output: key-value pairs like (Bus,1), (Car,1). Reduce Output: aggregated counts like (BUS,7), (CAR,7), (TRAIN,4).

Q. What is jps command used for in Hadoop?

Ans: jps (Java Virtual Machine Process Status) lists all running Java processes. In Hadoop, it is used to verify that Hadoop daemons (NameNode, DataNode, ResourceManager, etc.) are running.

Q. What are the main Hadoop daemons?

Ans: NameNode (manages HDFS metadata), DataNode (stores actual data blocks), ResourceManager (manages cluster resources in YARN), NodeManager (manages resources on individual nodes).

Q. What commands start and stop Hadoop?

Ans: start-dfs.sh starts HDFS daemons, start-yarn.sh starts YARN daemons, stop-all.sh stops all daemons.

Q. What is a JAR file in Hadoop MapReduce?

Ans: A JAR (Java ARchive) file packages all compiled Java classes and a Manifest file. It is submitted to Hadoop to run a MapReduce job using: `hadoop jar .`

Q. What is the Manifest.txt file in Hadoop?

Ans: Manifest.txt specifies the Main-Class (entry point/driver class) of the JAR file so Hadoop knows which class to invoke when running the job.

Assignment 2 — Log File Processing using MapReduce

Q. What is a distributed application?

Ans: A distributed application runs across multiple networked computers simultaneously, coordinating their tasks. Hadoop MapReduce is used to build distributed applications that process data in parallel.

Q. Why is MapReduce used for log file processing?

Ans: Log files can be extremely large (GBs to TBs). MapReduce processes them in parallel across a cluster, making analysis fast and scalable. Each Mapper processes a chunk of the log file independently.

Q. What is the role of the Driver class in Hadoop MapReduce?

Ans: The Driver class configures and submits the MapReduce job. It sets the Mapper class, Reducer class, input/output paths, and key/value types.

Q. What are the three main Java files in a MapReduce program?

Ans: Mapper.java (contains the Map logic), Reducer.java (contains the Reduce logic), and Driver.java (configures and submits the job).

Q. What is the `hdfs dfs -put` command used for?

Ans: It uploads files from the local filesystem to HDFS. Example: `hdfs dfs -put ~/input2000 /` copies the local input2000 folder to the HDFS root directory.

Q. What is the `hdfs dfs -cat` command?

Ans: `hdfs dfs -cat` is used to display the contents of a file stored in HDFS. Example: `hdfs dfs -cat /output2000/part-00000` prints the MapReduce job output.

Q. What is the `access_log_short.csv` file?

Ans: It is a sample web server access log file used as input for the log file analysis MapReduce job. It contains records of HTTP requests made to a server.

Q. What does the `javac` command do in Hadoop setup?

Ans: `javac` compiles Java source files (.java) into bytecode (.class files). The command includes `-d` to specify the output directory. Classpath must include Hadoop JARs.

Q. What is the Writable interface in Hadoop?

Ans: Writable is a Hadoop interface that enables serialization and deserialization of data types for transmission between nodes. All MapReduce key/value types must implement Writable.

Q. What is the WritableComparable interface?

Ans: WritableComparable extends Writable and Comparable. It is required for keys in MapReduce because keys must be sortable (for the shuffle/sort phase).

Assignment 3 — Weather Data Processing using MapReduce

Q. What is the aim of the Weather Data MapReduce assignment?

Ans: To locate a weather dataset (sample_weather.txt), read text input files using MapReduce, and compute the average temperature, dew point, and wind speed from the data.

Q. What is the major advantage of MapReduce?

Ans: The major advantage is easy scalability of data processing across multiple computing nodes. Large datasets can be processed in parallel without manual coordination.

Q. What are the three stages of MapReduce program execution?

Ans: Three stages: (1) Map Stage — processes input and produces intermediate key-value pairs, (2) Shuffle Stage — groups all values by key, (3) Reduce Stage — aggregates values for each key to produce final output.

Q. What is HDFS used for in the Weather Data assignment?

Ans: HDFS stores the weather dataset so MapReduce can distribute chunks across the cluster. The framework reads input from HDFS and writes output back to HDFS.

Q. How does MapReduce view input data?

Ans: The MapReduce framework views input as a set of key-value pairs and produces a set of key-value pairs as output. The types of input and output pairs can differ.

Q. What is the difference between Splitting and Shuffling in MapReduce?

Ans: Splitting divides the input data into fixed-size chunks sent to individual Mappers. Shuffling/Sorting groups all intermediate key-value pairs by key before sending them to Reducers.

Q. What is a Commodity Hardware cluster?

Ans: A cluster of inexpensive, standard (commodity) machines used together to process large data. Hadoop is specifically designed to run reliably on such hardware, handling node failures automatically.

Q. What is fault tolerance in Hadoop?

Ans: Hadoop achieves fault tolerance by replicating data blocks across multiple DataNodes (default 3). If a node fails, data can be retrieved from another replica.

Q. What is the difference between YARN and MapReduce?

Ans: YARN (Yet Another Resource Negotiator) manages cluster resources (CPU, memory) and job scheduling. MapReduce is the programming model that runs on top of YARN to process data.

Q. What is a NameNode and what happens if it fails?

Ans: The NameNode manages the HDFS file system namespace and metadata. If it fails (Single Point of Failure), the entire HDFS becomes inaccessible. Solutions include Secondary NameNode, Standby NameNode, or HDFS HA (High Availability).

GENERAL / Cross-Topic Viva Questions

Q. What is Big Data? What are the 3 V's?

Ans: Big Data refers to extremely large datasets that cannot be processed by traditional tools. The 3 V's are: Volume (huge amount of data), Velocity (speed of data generation), and Variety (different types: structured, semi-structured, unstructured).

Q. What is the difference between supervised and unsupervised learning?

Ans: Supervised learning uses labelled data (e.g., Linear Regression, Logistic Regression, Naïve Bayes). Unsupervised learning finds patterns in unlabelled data (e.g., K-Means Clustering, PCA).

Q. What is a dataset? What is a DataFrame?

Ans: A dataset is a structured collection of related data. A DataFrame is a 2-dimensional, tabular data structure in pandas (Python) with rows and columns, similar to a spreadsheet or SQL table.

Q. What is Scikit-learn (sklearn)?

Ans: Scikit-learn is a Python ML library that provides simple and efficient tools for data mining and ML, including algorithms for classification, regression, clustering, and model evaluation.

Q. What is the train-test split?

Ans: Train-test split divides the dataset into a training set (to build the model, typically 70-80%) and a test set (to evaluate performance, typically 20-30%). This helps assess how well the model generalizes.

Q. What is a feature and a label in ML?

Ans: A feature (independent variable) is an input attribute used to make predictions. A label (dependent variable) is the output the model tries to predict.

Q. What is Anaconda and Jupyter Notebook?

Ans: Anaconda is a Python distribution for scientific computing that includes libraries like NumPy, Pandas, and Scikit-learn. Jupyter Notebook is a web-based interactive environment for writing and running Python code.

Q. What is the difference between classification and regression?

Ans: Classification predicts a discrete/categorical output (e.g., spam/not spam). Regression predicts a continuous numeric output (e.g., house price).

Q. What are key Python libraries used in DSBDA practicals?

Ans: Pandas (data manipulation), NumPy (numerical operations), Matplotlib (plotting), Seaborn (statistical visualization), Scikit-learn (ML algorithms), NLTK (NLP).

Q. What is the difference between Hadoop and Spark?

Ans: Hadoop processes data on disk (slower but suitable for batch processing of very large data). Apache Spark processes data in-memory (much faster, suitable for real-time and iterative tasks). Spark can run on top of Hadoop.

End of DSBDA Viva Questions — All the Best for Your Examination!